



OSTIM TECHNICAL UNIVERSITY
ENGINEERING FACULTY
ELECTRICAL AND ELECTRONICS ENGINEERING

EEE308 PROJECT PROPOSAL

Sayme Ayça Öztekin (230202402)

İbrahim Yasin Aşıcı (230202042)

April 2026

Project Proposal Content List

1. Project Title
2. Project Objective
3. System Architecture
 - a. Microcontroller Unit (MCU)
 - b. Sensor Module
 - c. Display Module
 - d. Alert Module
 - e. Power Supply Module
 - f. Atmospheric Pressure Sensor Module
 - g. Prediction Module
4. Hardware Selection
5. Justification of Task Distribution

1. Project Title: Arduino Uno-Based Mini Weather Station with LCD Display and LED Alert

2. Project Objective

This project involves the design and implementation of a microcontroller based mini weather station using the Arduino Uno. This system measures ambient temperature and humidity using a DHT11 sensor. The microcontroller processes the received data and displays it on a 16x2 character LCD in real time. This system is not only storing data, but it also allows for simple temperature management. When the temperature exceeds(reaches) a specified threshold, LED lights up. The LED will remain ON. The main goal of this work is to show how to use basic concepts of embedded systems, such as connecting sensors, controlling digital input and output, and showing data in real time. In a wider perspective by taking all the scientific knowledge we learned during the theoretical sessions and applying it in a project to test our knowledge.

3. System Architecture

The system consists of several modules.

a. Microcontroller Unit (MCU)

We will use Arduino Uno. It works as the main control unit. The microcontroller controls all system functions, such as sensor data collecting, processing, display control, and alert mechanism for decision making processes.

b. Temperature and Humidity Sensor Module

The sensor module consists of the DHT11 temperature and humidity sensor. This sensor sends digital data over a single wire. A single wire communication protocol is a way to send data between devices over a single data line using very accurate timing signals. The sensor measures the temperature and humidity of the surrounding environment and sends the information to the microcontroller for additional processing.

c. Display Module

The display module is implemented using a 16x2 character LCD. The LCD works in 4 bit parallel mode which reduces the number of I/O pins needed. It shows the sensor's real time temperature and humidity readings.

d. Alert Module

The alert module contains an LED and a current limiting resistor. This module has a visual indicator that shows when the observed temperature goes above a set level.

e. Power Supply Module

We will use the MB102 breadboard power module that supplies the system and gives a steady 5V supply. This makes sure that the microcontroller, sensor, and display units all work well together. Also, it gives us flexibility to work on that project outside of the lab.

f. Atmospheric Pressure Sensor Module

The pressure sensing module measures atmospheric pressure using the BMP280 sensor and provides this data to the microcontroller. Pressure is an important parameter for interpreting weather conditions. The BMP280 communicates with the Arduino Uno via the I2C or SPI interface, enabling efficient digital data transfer. The measured pressure values are processed by the microcontroller and displayed on the LCD. Additionally, this data is used as an input for the simplified rain probability prediction module. Although the BMP280 can also measure temperature, in this project it is used only for pressure measurement, while temperature and humidity data are obtained from the DHT11 sensor.

g. Prediction Module

The prediction module is designed to estimate the probability of rainfall using a simplified rule-based model based on three measured environmental parameters: temperature, relative humidity, and atmospheric pressure. The primary objective of this module is to provide a real-time, sensor-based estimation of rain likelihood without introducing complex meteorological calculations. In this implementation, temperature and humidity data are obtained from the DHT11 sensor, while atmospheric pressure is measured using the BMP280 sensor. These three parameters are processed directly within the microcontroller to generate a numerical estimation of rain probability. The rain probability is calculated using a weighted combination of the three parameters, where humidity provides the dominant contribution, atmospheric pressure represents weather stability, and temperature acts as a secondary adjustment factor. The resulting value is then normalized to a percentage scale between 0% and 100% to ensure a clear and user-friendly output. High relative humidity indicates that the air is closer to saturation, low atmospheric pressure suggests unstable weather conditions, and lower temperatures can contribute to condensation processes. These physical interpretations form the basis of the simplified prediction model. The output of the prediction module is displayed on the LCD allowing the user to easily interpret current weather conditions.

4. Hardware Selection

We will use the components that we purchase.

- Arduino Uno
- DHT11
- BMP280
- I2C or parallel LCD
- LED
- 2 or 3 × 100nF Ceramic Capacitor

- 220 Ω
- breadboard

Our main purpose is using the lab provided components but such a failure condition these will be purchased by our side.

5. Justification of Task Distribution

Sayme Ayça Öztekin is in charge of Power Supply Module, Display Module, Alert Module, Atmospheric Pressure Sensor Module . İbrahim Yasin Aşıcı is in charge of Microcontroller Unit (MCU) and Sensor Module and Prediction Module. Every group member is responsible for their report writing parts that done in practically and If any member is unable to perform their assigned duties, we can make little job changes among ourselves.

In that task distribution process, we focused on delivering as comprehensive and equitable work as possible to remain as focused as possible.